Organism Similarity: Is There More Genetic Difference Within Yeast Than Within Mammals?

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Yes, at the genus and the subphylum, which is where the claim is made (Peris et al. 2023, Fig. 3c), and the data here reproduce it (terms in Table 1). Within *Saccharomyces*, S288C/*S. paradoxus* AAI is 92.1% and S288C/*S. eubayanus* 83.5%, so the genus spans the distance from human/macaque (95.3%) to human/mouse (83.7%). The median of the 75 Saccharomycotina pairs outside the genus is 47.4% AAI, below every one of the 36 vertebrate pairs (lowest 71.5%). Within *S. cerevisiae*, an isolate's typical ORF is 99.72% identical to S288C at the protein level (median over the 1,011 isolates of each isolate's per-ORF median), and 98.04% when every frameshifted or indel-bearing ORF is charged in full; one human genome differs from GRCh38 at 99.90 to 99.91% protein identity, a literature value not computed here. Measured: AAI and ortholog fraction over reciprocal best mmseqs2 hits for the 528 pairs of 33 per-gene proteomes, EC-4 repertoires from one eggNOG-mapper run, isolate protein identity from Peter et al. 2018's reference-inferred alleles, and the clade structure of Peter's SNP distance matrix. For carrying a model trained on the isolates to *P. kudriavzevii* SD108, the numbers that matter are Figure 1's: SD108 is 47.1% AAI from S288C, beyond every vertebrate pair, and 59.5% of its genes have a one-to-one ortholog in S288C.

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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1 Controlled vocabulary ✗

Table 1: Terms used in the figures and tables, in alphabetical order.

Term	Definition
AAI	Average amino-acid identity: the mean percent identity over the reciprocal best hits of two per-gene proteomes (mmseqs2 local alignments, `03_mmseqs_rbh.py`).
Amniote	In this organism set, a mammal or the chicken.
EC-4 Jaccard	Jaccard index of the sets of complete, four-level EC numbers that one eggNOG-mapper run assigns to the two proteomes (`04_emapper.py`).
Ortholog fraction	Reciprocal best hits over the gene count of the smaller per-gene proteome, $n_{\mathrm{RBH}}/\min(n_i, n_j)$: the share of the smaller organism's genes that have a one-to-one ortholog in the other.
Ortholog Jaccard	Reciprocal best hits over the union of the two per-gene proteomes, $n_{\mathrm{RBH}}/(n_i + n_j - n_{\mathrm{RBH}})$.
Per-gene proteome	One protein per gene, the longest, keyed by NCBI GeneID where the annotation carries it (`02_proteome_prep.py`).
Peter clade	The clade an isolate carries in Peter et al. 2018's Supplementary Table 1, with the Wine/European subclades merged; isolates the table gives no clade form one more group.
Reciprocal best hit (RBH)	A pair of proteins, one from each proteome, each the other's best mmseqs2 hit; the proxy for a one-to-one ortholog.
Single-copy OG proxy	AAI restricted to Eukaryota-level eggNOG orthologous groups that hold exactly one protein in each member of the pair; stands in for Peris et al. 2023's BUSCO-restricted AAI.
SNP distance	Percent non-identical bases between two isolates in Peter et al. 2018's released distance matrix, heterozygous differences half-weighted.

2 Figures ✗

SRC: 07_figures.py fig6_main_transfer_ladder; s288c_centered.csv

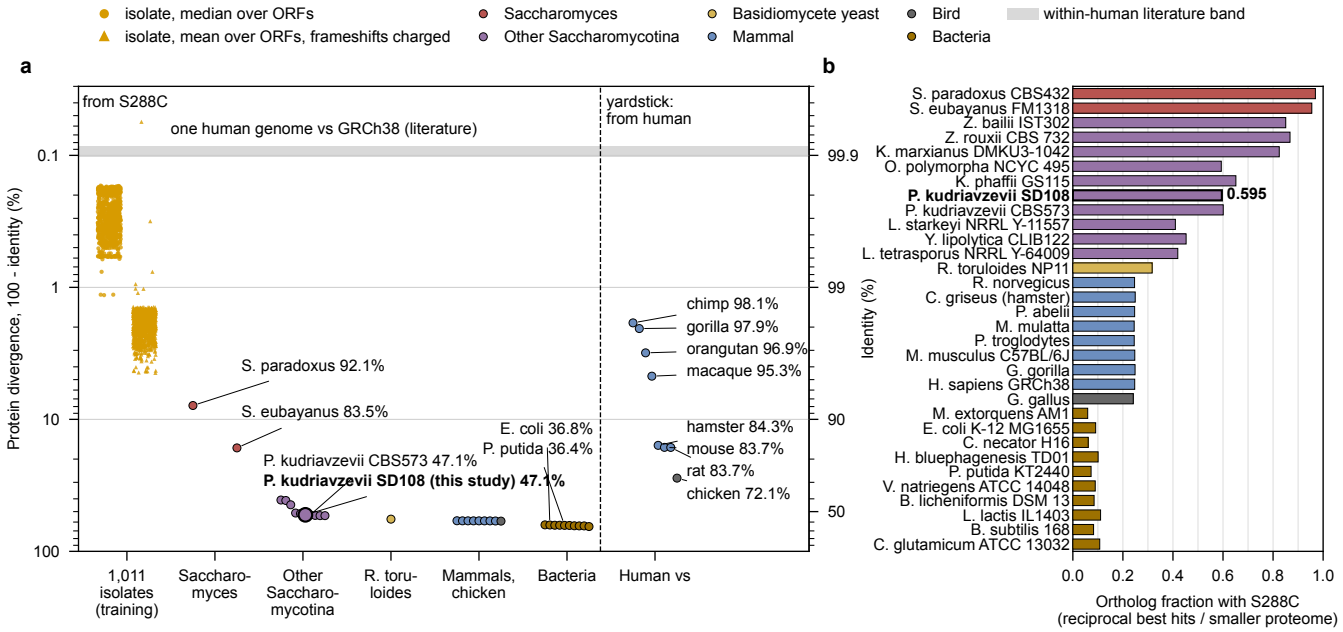


Figure 1: The training isolates reach human/chimp protein divergence from S288C; *P. kudriavzevii* SD108 lies beyond every vertebrate pair, with 59.5% of its genes in one-to-one orthology with *S. cerevisiae*. (a) Protein divergence from S288C (100 minus identity, log scale, inverted; identity on the right), columns ordered by distance, colored by the partner's taxonomic group (legend). SD108 (large point, black edge) is 47.1% AAI from S288C, below the lowest of the 36 vertebrate pairs (71.5%; human/chicken 72.1%). The 1,011 isolates appear twice: circles are each isolate's median over ORFs of protein identity to S288C (median over isolates 99.72%), triangles its mean over ORFs, which charges every gap column of a frameshifted or indel-bearing ORF (median 98.04%, lowest 95.55%, against human/chimp AAI 98.14%). The other 10 Saccharomycotina hosts sit at 46.4 to 59.2% AAI, the 9 mammals and chicken at 41.0 to 41.4%, the 10 bacteria at 35.1 to 37.0%. Right of the dashed line, the same axis measured from human over 8 partners, as a yardstick (mammals blue, chicken gray). Gray band: one human genome vs GRCh38 at peptide-altering sites (1000 Genomes Project 2015; literature value). (b) Ortholog fraction with S288C for the organisms of (a) without the isolates, same order and colors. SD108 shares 3,072 reciprocal best hits with S288C, 59.5% of its 5,159 genes; the other Saccharomycotina range from 0.410 to 0.868, and no bacterium exceeds 0.111.

SRC: 07_figures.py fig1_claim_protein_divergence

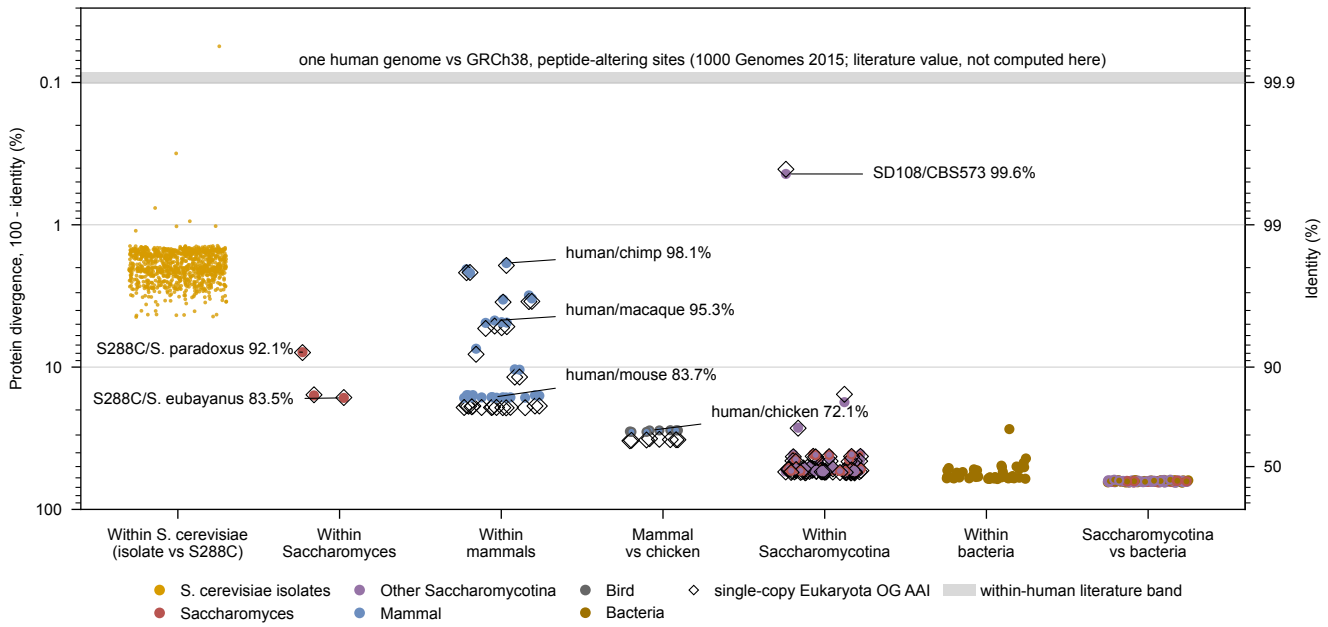


Figure 2: *Saccharomyces* reaches the human/mouse distance, and the median Saccharomycotina pair lies beyond human/chicken. Each point is a pair; the y axis is 100 minus protein identity (log scale, inverted; identity on the right). Within *Saccharomyces* AAI runs from 83.5 to 92.1% (3 pairs), within mammals from 83.5 to 98.1% (28 pairs), mammal vs chicken 71.5 to 72.2% (8); the 75 Saccharomycotina pairs outside the genus have a median of 47.4%, the 45 bacterial pairs 40.5% and the 130 Saccharomycotina–bacteria pairs 36.5%. Within *S. cerevisiae* (1,011 isolates), each point is the mean over S288C ORFs of an isolate's protein identity to S288C with every gap column charged; the per-isolate median over ORFs gives a median divergence of 0.28%. Every other column is AAI over reciprocal best hits. Open diamonds: the single-copy OG proxy. Point face and edge colors are the taxonomic groups of the two members, so a within-group pair is one solid color and a cross-group pair carries both. Gray band: one human genome differs from GRCh38 at 10,000 to 12,000 peptide-altering sites (1000 Genomes Project 2015; literature value), divided by the 11.7 million residues of the human per-gene proteome.

SRC: 07_figures.py fig2_cladogram_heatmaps

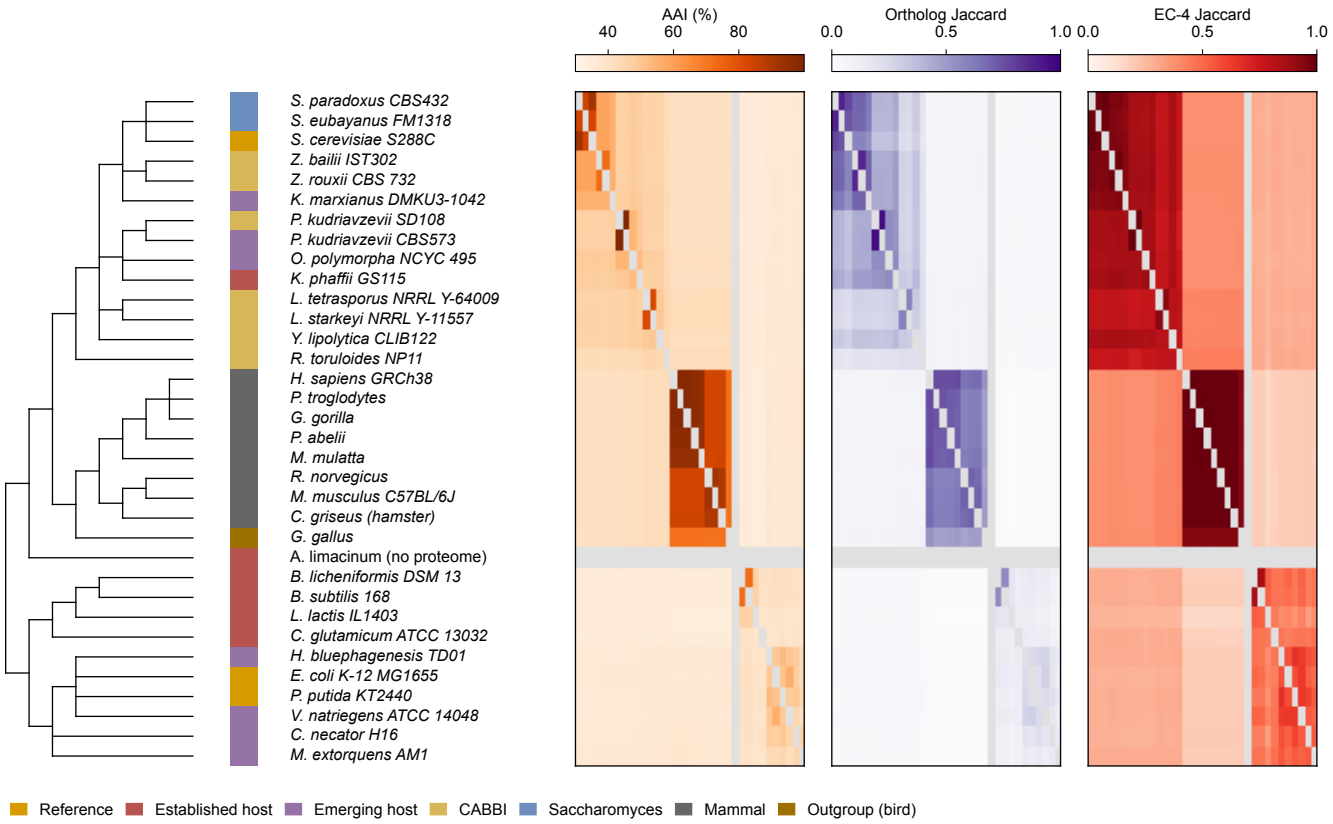


Figure 3: The 34 organisms on three similarity axes, in NCBI taxonomy order. Left: NCBI taxonomy cladogram (topology only; ete3 NCBITaxa, `05_trees.py`); the color strip is each organism's first group in the experiment plan (legend: reference, established host, emerging host, CABBI, *Saccharomyces*, mammal, outgroup). Heatmaps, rows and columns in cladogram order: AAI, ortholog Jaccard and EC-4 Jaccard (eggNOG-mapper on eggNOG 5.0.2, `04_emapper.py`); color scales above each map. Gray: no proteome (*A. limacinum* has no annotated assembly at NCBI) or the diagonal.

SRC: 07_figures.py fig3_taxonomy_vs_aai_nj

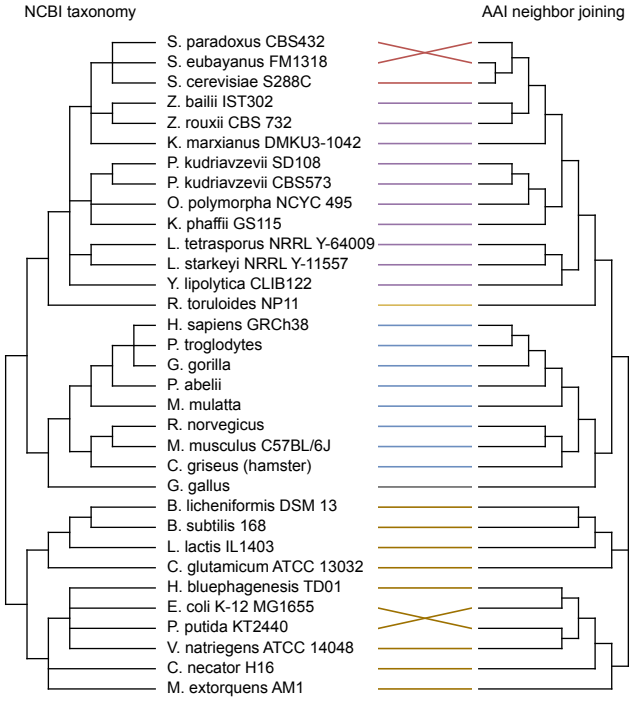


Figure 4: NCBI taxonomy and AAI agree on the large groups. Left: NCBI taxonomy cladogram (topology only) over the 33 organisms with a proteome. Right: neighbor-joining tree (Biopython) on $d = -\ln(\text{AAI}/100)$, rooted on the bacteria, drawn as topology with children rotated to follow the left tree. Lines join the same organism and take its taxonomic group's color (`05_trees.py`).

SRC: 07_figures.py fig4_clade_tree_within_distance; clade_snp_distance.csv, clades_nj.nwk, clade_distance_summary.csv

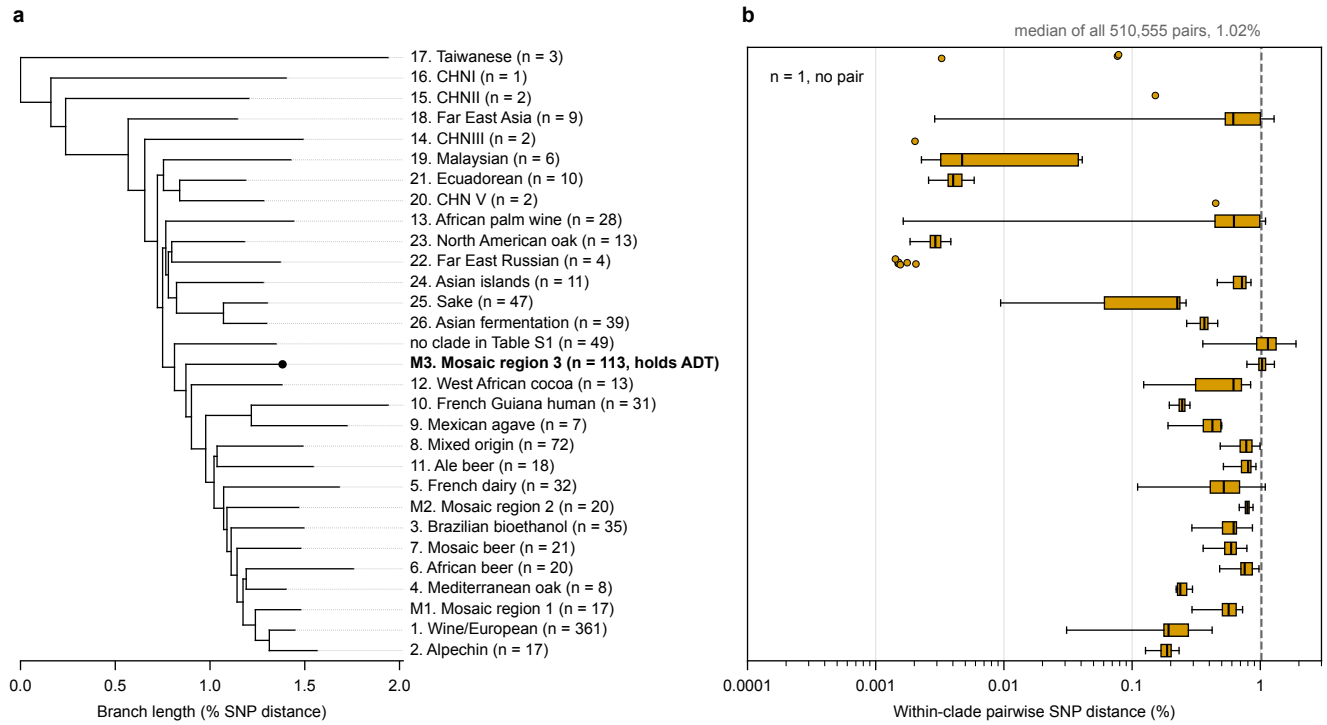


Figure 5: Every named *S. cerevisiae* clade but one is tighter than the species as a whole. (a) Neighbor-joining tree (Biopython) over 30 groups of Peter et al. 2018's 1,011 isolates: the 29 Peter clades and the 49 isolates with no clade. The distance between two groups is the mean SNP distance over every pair of their members; midpoint rooted, branch lengths in percent. Bold leaf and black dot: the clade holding ADT, the isolate with the lowest mean per-ORF nucleotide divergence from S288C in experiment 018 (0.0005%); S288C itself is not in Peter's matrix. (b) SNP distance between every pair of isolates within a group, same rows. 27 of the 28 named clades with at least one pair have a median below the median over all 510,555 isolate pairs (1.02%, dashed gray line); the exception is M3. Mosaic region 3 (1.03%). Clade medians start at 0.0015% (22. Far East Russian), and their median is 0.44%. The isolates with no clade, a mixed group, have a median of 1.15%. Boxes (groups with at least 10 pairs): median, quartiles and whiskers at one and a half times the interquartile range, outliers not drawn; points: every pair of a smaller group. One color for every group, since color carries no category here. Log axis.

SRC: 07_figures.py fig5_ortholog_fraction_vs_aai

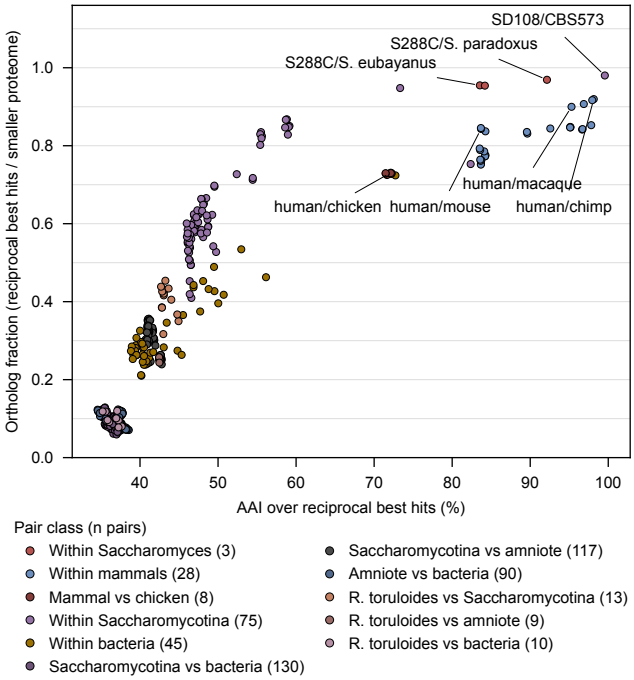


Figure 6: Shared orthologs fall with AAI. Each point is one of the 528 pairs of the 33 proteomes: AAI against the ortholog fraction (`03_mmseqs_rbh.py`). Color is the pair class, 11 in all (n pairs in the legend): within *Saccharomyces* 3, within mammals 28, mammal vs chicken 8, within Saccharomycotina 75, within bacteria 45, Saccharomycotina vs bacteria 130, Saccharomycotina vs amniote 117, amniote vs bacteria 90, *R. toruloides* vs Saccharomycotina 13, vs amniote 9 and vs bacteria 10. A within-group class takes that group's color in Figure 2; a cross-group class takes a palette color no group uses. Points are drawn largest class first, and the cross-kingdom pairs at the lowest AAI overlap.

3 Headline numbers ✗

Table 2: Headline numbers. Script column: the numbered script under `experiments/044-organism-similarity/scripts/` that computed the value; every value is read from `results/headline_numbers.json` or `results/figure_stats.json`.

Quantity	Value	Script
Isolate vs S288C protein identity, median over isolates of per-ORF median	99.72% (n = 1,011 isolates, 6,005 ORFs each)	06
Same, median over isolates of mean over ORFs	98.04% (range 95.55 to 99.94%)	06
Same, mean over equal-length ORFs only, median over isolates	99.58%	06
One human genome vs GRCh38, peptide-altering sites (literature)	10,000 to 12,000, i.e. 99.90 to 99.91% identity	00
human/chimp AAI, mean / median over RBHs	98.14 / 99.20% (n_{RBH} 18,461; ortholog fraction 0.920)	03
human/macaque AAI, mean / median over RBHs	95.29 / 96.90% (n_{RBH} 18,064; ortholog fraction 0.900)	03
human/mouse AAI, mean / median over RBHs	83.70 / 86.50% (n_{RBH} 16,925; ortholog fraction 0.843)	03
human/chicken AAI, mean / median over RBHs	72.12 / 73.60% (n_{RBH} 13,170; ortholog fraction 0.731)	03
S288C/*S. paradoxus* AAI, mean / median over RBHs	92.15 / 93.20% (n_{RBH} 5,367; ortholog fraction 0.969)	03
S288C/*S. eubayanus* AAI, mean / median over RBHs	83.55 / 84.70% (n_{RBH} 5,136; ortholog fraction 0.955)	03
SD108/CBS573 AAI, mean / median over RBHs	99.56 / 99.80% (n_{RBH} 5,037; ortholog fraction 0.980)	03
S288C/*E. coli* AAI	36.84% (n_{RBH} 391; ortholog fraction 0.091)	03
S288C/*P. kudriavzevii* SD108 AAI	47.1% (n_{RBH} 3,072 of 5,159 SD108 genes; ortholog fraction 0.595)	03
Within *Saccharomyces* (3 pairs)	AAI median 84.2%, range 83.5 to 92.1%	07
Within mammals (28 pairs)	AAI median 84.3%, range 83.5 to 98.1%	07
Mammal vs chicken (8 pairs)	AAI median 72.1%, range 71.5 to 72.2%	07
Within Saccharomycotina (75 pairs)	AAI median 47.4%, range 46.0 to 99.6%	07
Within bacteria (45 pairs)	AAI median 40.5%, range 38.8 to 72.7%	07
Saccharomycotina vs bacteria (130 pairs)	AAI median 36.5%, range 35.1 to 38.2%	07
Single-copy OG AAI vs RBH AAI, 253 eukaryote pairs	Pearson r 0.998; proxy lower by 1.66 points at the median	06, 07
Within-clade SNP distance, 28 named clades with a pair	clade medians 0.0015 to 1.03%; all 510,555 isolate pairs, median 1.02%	07

4 Organisms ✗

Table 3: The 34 organisms, 33 with a proteome. Source: *tier* is a torchcell genomes tier set, *NCBI* an assembly fetched with `datasets`; *Per gene* is the protein count after keeping the longest protein per gene (`02_proteome_prep.py`; `results/organisms.json`).

Organism	Source	Accession or tier set	Strain	Per gene
S. cerevisiae S288C	tier	sgd_S288C_R64-4-1_20230830	S288C	6,716
P. kudriavzevii SD108	tier	pkudriavzevii_SD108_v1	SD108	5,159
P. kudriavzevii CBS573	NCBI	GCF_003054445.1	CBS573	5,139
E. coli K-12 MG1655	tier	GCA_000005845.2	K-12 MG1655	4,290
P. putida KT2440	tier	GCA_000007565.2	KT2440	5,564
C. glutamicum ATCC 13032	NCBI	GCF_000011325.1	ATCC 13032	2,941
B. subtilis 168	NCBI	GCF_000009045.1	168	4,237
B. licheniformis DSM 13	NCBI	GCF_034478925.1	ATCC 14580	4,169
K. phaffii GS115	NCBI	GCF_000027005.1	GS115	5,040
C. griseus (hamster)	NCBI	GCF_003668045.3	17A/GY	21,776
H. sapiens GRCh38	NCBI	GCF_000001405.40	none	20,076
L. lactis IL1403	NCBI	GCF_000006865.1	IL1403	2,230
A. limacinum	absent	GCA_036850685.1 (no annotation)	none	none
Y. lipolytica CLIB122	NCBI	GCF_000002525.2	CLIB122	6,471
R. toruloides NP11	NCBI	GCF_000320785.1	NP11	8,140
C. necator H16	NCBI	GCF_000009285.1	H16	6,618
K. marxianus DMKU3-1042	NCBI	GCF_001417885.1	DMKU3-1042	4,941
H. bluephagenesis TD01	NCBI	GCF_054790765.1	TD01	3,635
O. polymorpha NCYC 495	NCBI	GCF_001664045.1	NCYC 495 leu1.1	5,153
V. natriegens ATCC 14048	NCBI	GCF_001456255.1	ATCC 14048	4,504
M. extorquens AM1	NCBI	GCF_000022685.1	AM1	6,151
L. starkeyi NRRL Y-11557	NCBI	GCA_001661325.1	NRRL Y-11557	8,182
Z. bailii IST302	NCBI	GCA_900162805.1	IST302	5,141
Z. rouxii CBS 732	NCBI	GCF_000026365.1	CBS 732	4,991
L. tetrasporus NRRL Y-64009	NCBI	GCF_029532465.1	NRRL Y-64009	8,004
S. paradoxus CBS432	NCBI	GCF_002079055.1	CBS432	5,538
S. eubayanus FM1318	NCBI	GCF_001298625.1	FM1318	5,379
P. troglodytes	NCBI	GCF_028858775.2	AG18354	22,543
G. gorilla	NCBI	GCF_029281585.2	KB3781	22,640
P. abelii	NCBI	GCF_028885655.2	AG06213	22,203
M. mulatta	NCBI	GCF_049350105.2	MMU2019108-1	21,513
M. musculus C57BL/6J	NCBI	GCF_000001635.27	C57BL/6J	22,198
R. norvegicus	NCBI	GCF_036323735.1	BN/NHsdMcwi	23,209
G. gallus	NCBI	GCF_016699485.2	bGalGal1	18,023

5 What was not done ✗

- **BUSCO itself.** The BUSCO-restricted AAI is the single-copy OG proxy; compleasm and a current busco were not installable in the experiment's environment.
- ***A. limacinum*.** No annotated assembly exists at NCBI for *Aurantiochytrium limacinum* or *Schizochytrium*; the organism stays on the cladogram (Figure 3) and is excluded from every proteome axis.
- ***Z. bailii* CLIB 213.** Its GenBank annotation is partial, so the same species' strain IST302, with a complete annotation, stands in; a strain substitution, not a species one, and the figure labels say IST302.
- **CLEAN for EC annotation** is not run; eggNOG-mapper is the annotator.
- **Isolates through the mmseqs2 RBH pipeline.** Isolate identity comes from a global alignment that charges every gap column, species AAI from local alignments of reciprocal best hits; the isolate proteomes were not run through the species pipeline.
- **One AAI-matched isolate statistic.** The mean and the median over ORFs are both reported and bracket the comparison with species AAI; neither is the same statistic as AAI.